

Q.25 Define modulus of elasticity. Explain

- a) Young's modulus of elasticity
- b) Bulk's modulus of elasticity

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**1st Year / Textile Design.**  
**Subject : Applied Sciences**

Time : 3 Hrs.

M.M. : 60

**SECTION-A**

**Note:** Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Smallest unit of electric current is :

- a) Volt
- b) Ampere
- c) Columb
- d) None

Q.2 An aqueous solution whose pH = 2.4 is.

- a) Acidic
- b) Neutral
- c) Basic
- d) Amphoteric

Q.3 S.T. Unit of power is

- a) Volt
- b) Ampere
- c) Watt
- d) Columb

Q.4 Pressure in fluids is equal to

- a)  $hdg$
- b)  $mgh$
- c)  $mcl$
- d)  $mc^2$

- Q.5 Formula of Kinetic energy is.  
 a)  $mgh$                               b)  $\frac{1}{2} mcl^2$   
 c)  $mcl$                                  d)  $V^2 - \mu^2 = 2as$
- Q.6 Which of the following is not unit of temperature.  
 a) Kelvin                                b) Fahrenheit  
 c) Celcius                               d) Volt

### SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Anions are \_\_\_\_\_ charged loins.  
 Q.8 Oxidations is a process of loss electrons. (True/False)  
 Q.9 pH of blood is around \_\_\_\_.  
 Q.10 Define Power.  
 Q.11 Give an example of Positive work.  
 Q.12 S.I. Unit of Stress is \_\_\_\_.

### SECTION-C

**Note:** Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Differentiate between Conductor and Insulation.

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- Q.14 Define Molarity and Normality.  
 Q.15 Explain Faraday's first law of electrolysis.  
 Q.16 Define stress and strain.  
 Q.17 What are the principles for measurement of temperature?  
 Q.18 Define term potential energy and Kinetic energy.  
 Q.19 Differentiate between Atmospheric pressure and absolute pressure.  
 Q.20 What are the effects of temperature on surface tension?  
 Q.21 What is meant by pressure? Write units and dimensional formula.  
 Q.22 Explain Electroplating.

### SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain pH and its applications.  
 Q.24 Explain the law of conservation of energy for free falling body, show that mechanical energy remains same.

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